What is Selenium?

Selenium is a free (open-source) automated testing framework used to validate web applications across different browsers and platforms. You can use multiple programming languages like Java, C#, Python, etc to create Selenium Test Scripts. Testing done using the Selenium testing tool is usually referred to as Selenium Testing.

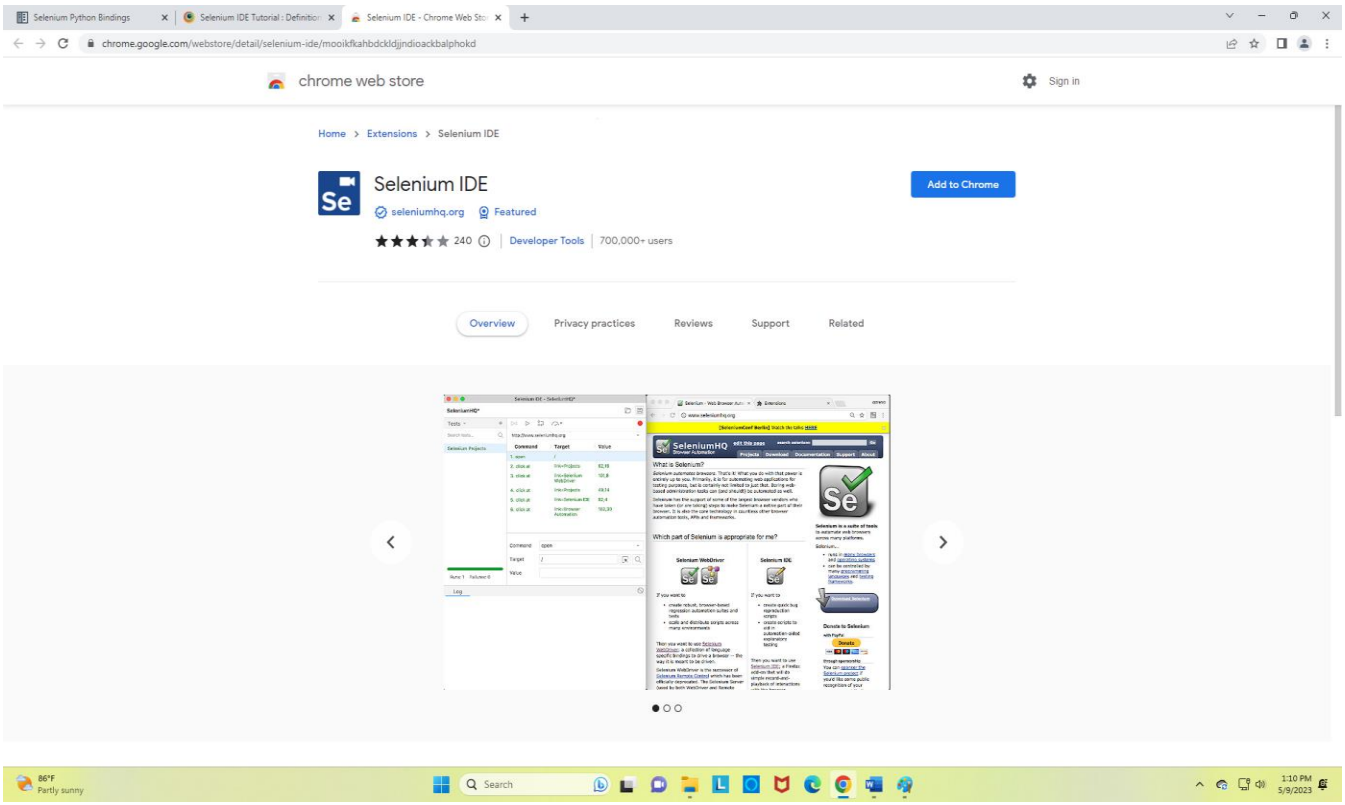
Who developed Selenium?

Selenium was created by **Jason Huggins in 2004**. An engineer at ThoughtWorks, he was working on a web application that required frequent testing.

What is Selenium IDE?

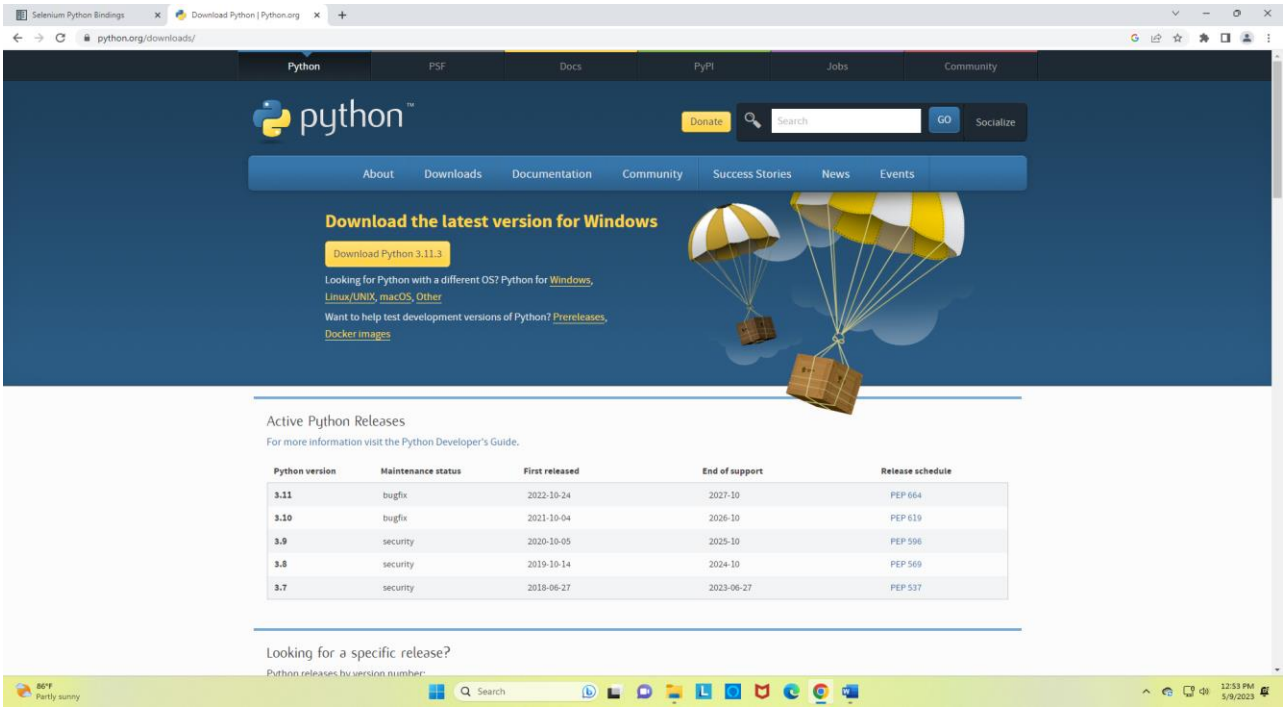
Selenium IDE (Integrated Development Environment) is primarily a record/run tool that a test case developer uses to develop Selenium Test cases. Selenium IDE is an easy-to-use tool from the Selenium Test Suite and can even be used by someone new to developing automated test cases for their web applications. One does not require any special setup to get started with Selenium IDE. You just need to add the extension of your specific browser. Selenium IDE provides you with a GUI (Graphical User Interface) for easily recording your interactions with the website.

Selenium IDE allows a user or a test case developer to create the test cases and test suites and edit it later as per their requirements. The development environment also provides the capability of converting test cases to different programming languages, which makes it easier for the user and does not mandate the need for knowing a specific programming language.



Installing Python bindings for Selenium

1) **First download and install python**



2) **Open cmd prompt and install selenium**

Use pip to install the selenium package. Python 3 has pip available in the standard library. Using pip, you can install selenium like this: **pip install selenium**

3) **Download web browser drivers**

What is Chrome Driver?

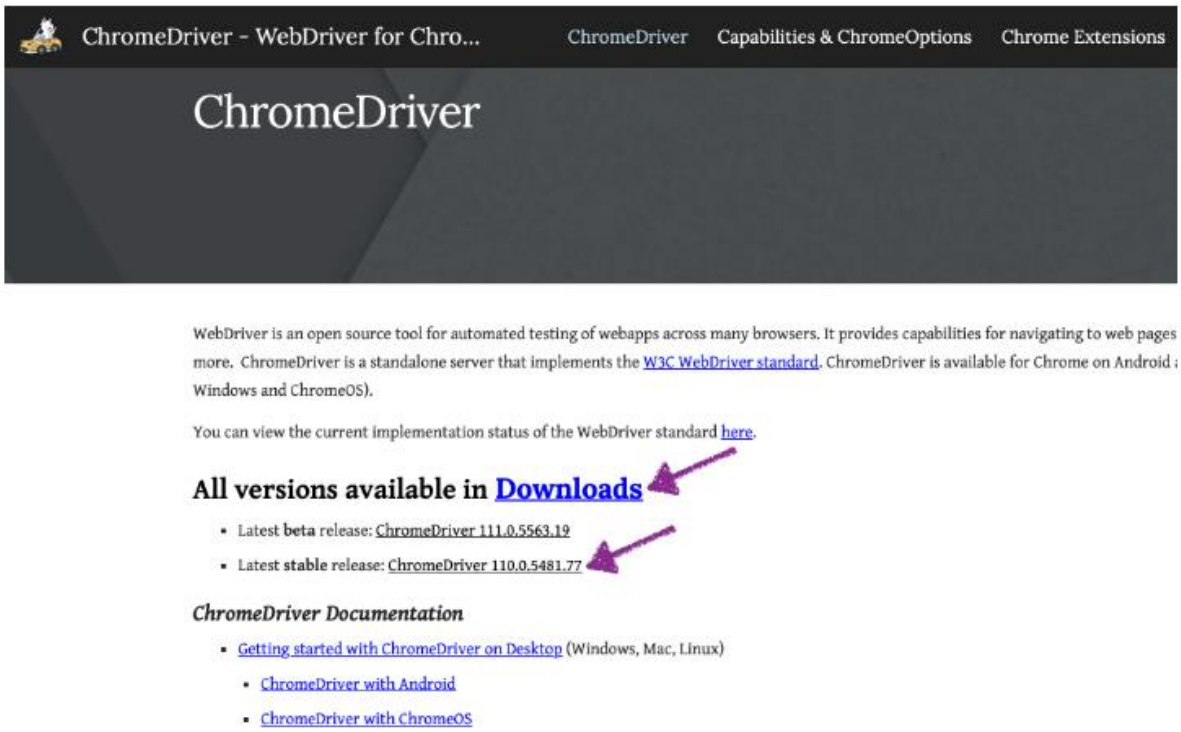
You know that selenium is a tool that basically interacts with browsers to test your web applications. you can use selenium webdriver to open some url on a browser and interact with webpage elements like buttons, links, text boxes etc.

Steps to download ChromeDriver

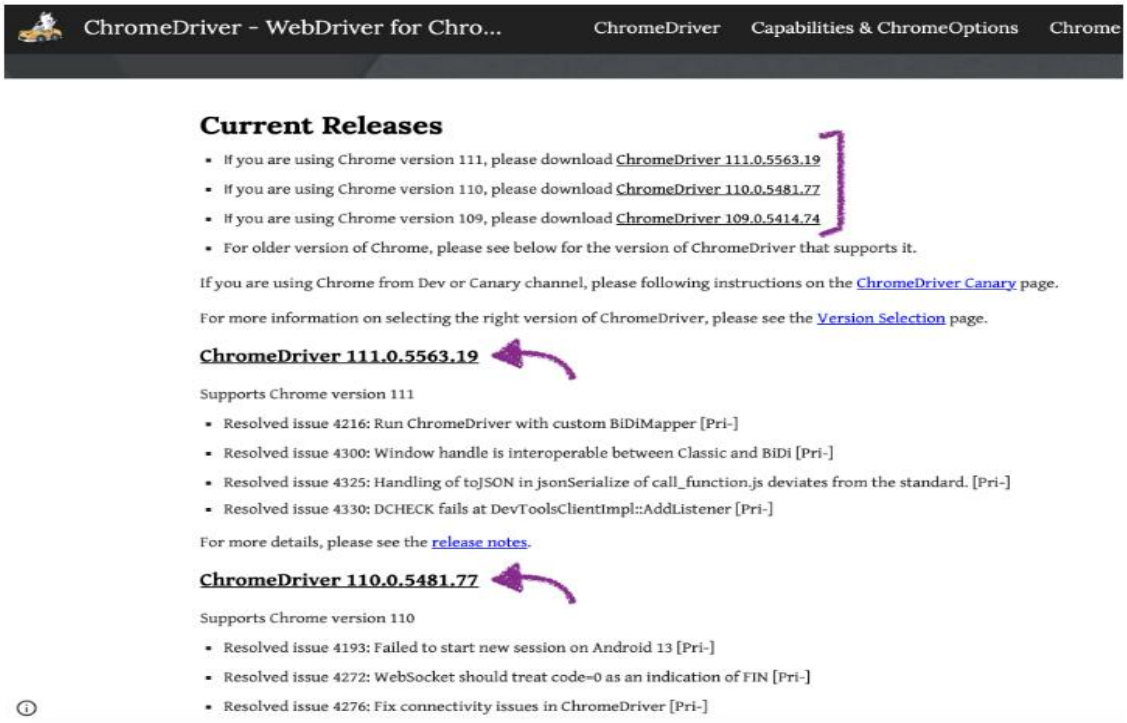
Follow the steps given below to download the latest version of chrome driver for selenium –

1. Open ChromeDriver page – https://chromedriver.chromium.org/home

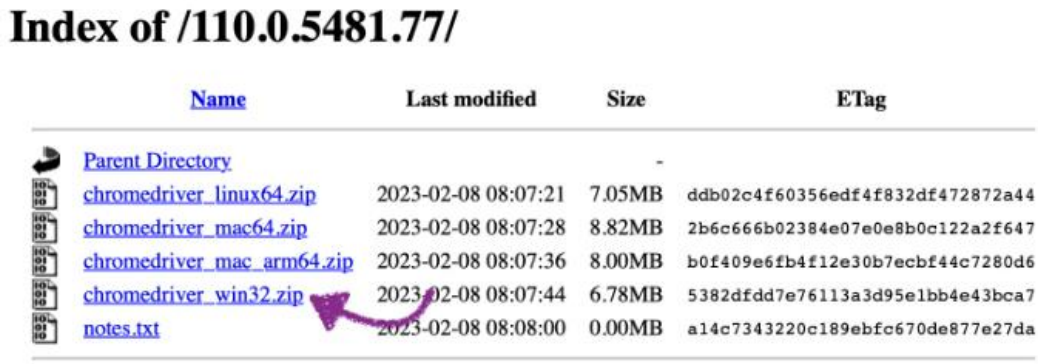
2. This page provides the latest beta and stable ChromeDriver release information



3. Click on Downloads link. Based on the version of Chrome browser you have on your machine, click on the corresponding ChromeDriver version selenium-chromedriver-latest-version



4. Assuming you clicked on the link for ChromeDriver 110, you will be navigated to ChromeDriver download page which contains ChromeDriver for Mac, Windows and Linux operating systems.



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Name	Last modified	Size	ETag
Parent Directory		-	
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chromedriver_mac_arm64.zip	2023-02-08 08:07:36	8.00MB	b0f409e6fb4f12e30b7ecbf44c7280d6
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notes.txt	2023-02-08 08:08:00	0.00MB	a14c7343220c189ebfc670de877e27da

chromedriver-for-windows-and-other-os
5. Click on chromedriver_win32.zip to download ChromeDriver for Windows.
6. Once you download the zip file, unzip it to retrieve chromedriver.exe

With this, we complete the download process for ChromeDriver.

The first thing you'll want to do with WebDriver is navigate to a link. The normal way to do this is by calling get
method:
driver.get("http://www.google.com")
WebDriver will wait until the page has fully loaded (that is, the onload event has fired) before returning control to
your test or script. Be aware that if your page uses a lot of AJAX on load then WebDriver may not know when it has
completely loaded. If you need to ensure such pages are fully loaded then you can use waits.

Writing test script with python

```
 from selenium import webdriver
from selenium.webdriver.common.keys import Keys
import time
driver = webdriver.Chrome(executable_path= "f:\\spl\\chromedriver.exe")  #create object
# get
driver.get("https://www.google.com/")
print(driver.title)
time.sleep(5)
driver.get("https://www.amazon.in/")
print(driver.title)
driver.back()
print(driver.title)
driver.forward()
driver.quit()
```